

Class XI | Unit 2 | weight 10%

Computer memory

Board weight about 10% | 10 theory periods

Memory is where the computer holds bits. This lecture teaches the units used to measure that holding, then the chips the CPU uses, then the devices that keep work after the computer is switched off.

What you should be able to do

- Use bit, nibble, byte, kilobyte, megabyte, gigabyte and terabyte correctly.
- Distinguish volatile and non-volatile memory.
- Explain SRAM, DRAM, ROM, PROM, EPROM and EEPROM.
- Describe hard drives, USB flash drives, external hard drives, memory cards and combo drives.

Path through the lecture

- 1 Practise the unit ladder with two numerical conversions before naming chips.
- 2 Teach volatile and non-volatile with the power cable as the test.
- 3 Compare SRAM with DRAM, then the ROM family, in tables students can reproduce.
- 4 Finish with secondary devices they can point to in the lab.

2.1 Units of memory

A **bit** is one binary digit, 0 or 1. It is the smallest unit of storage. A **nibble** is 4 bits. A **byte** is 8 bits and is the usual unit for one character in the older character codes.

The ladder used in this course

Unit	Equals
1 nibble	4 bits
1 byte	8 bits
1 kilobyte (KB)	1024 bytes
1 megabyte (MB)	1024 KB
1 gigabyte (GB)	1024 MB
1 terabyte (TB)	1024 GB

In this course the multiplier is 1024, which is 2 to the power 10. Worked example: a file of 2 KB contains $2 \times 1024 = 2048$ bytes, and $2048 \times 8 = 16384$ bits. Worked example the other way: 4096 MB is $4096 / 1024 = 4$ GB.

Exam point

Write the unit in the answer. A bare 2048 is not a complete conversion if the question asked for bytes.

2.2 Primary memory

Volatile memory. Memory that loses its stored contents when the power supply is removed.

Non-volatile memory. Memory that keeps its stored contents when the power supply is removed.

RAM is volatile working memory. ROM and secondary storage are non-volatile. The test is simple: switch the power off and ask whether the contents are still there.

RAM

RAM. Random access memory. The processor can reach any location directly, without reading all the earlier locations. It holds the program and data that are in use, and it is volatile.

Static RAM (SRAM) stores each bit in a small circuit of transistors arranged as a flip-flop. It does not need a refresh. It is fast and costly, so it is used in small amounts as cache inside or beside the processor.

Dynamic RAM (DRAM) stores each bit as a charge in a tiny capacitor. The charge leaks, so the memory must be refreshed many times a second. DRAM is slower than SRAM and much cheaper per bit, so the large main memory of a PC is DRAM.

SRAM and DRAM

Basis	SRAM	DRAM
Cell	Flip-flop made of transistors	Capacitor and transistor
Refresh	Not required	Required
Speed	Faster	Slower
Cost and density	Costlier, fewer bits on a chip	Cheaper, more bits on a chip
Typical use	Cache	Main memory

ROM and its family

ROM. Read-only memory. In ordinary work the processor reads it and does not rewrite it. It is non-volatile and often holds the start-up program.

- **PROM**, programmable ROM, is written once by the user with a special device. After that the contents stay fixed.
- **EPROM**, erasable programmable ROM, can be erased by ultraviolet light through a window on the chip and then programmed again.
- **EEPROM**, electrically erasable programmable ROM, is erased and rewritten by electrical signals. A location can be updated without removing the chip. Flash memory, used in pen drives and SSDs, is a form of EEPROM erased in blocks.

The ROM family

Type	Written	Erased
ROM	At manufacture	Not erasable in ordinary use
PROM	Once, by the user	Not erasable
EPROM	By a programmer	By ultraviolet light
EEPROM	Electrically	Electrically

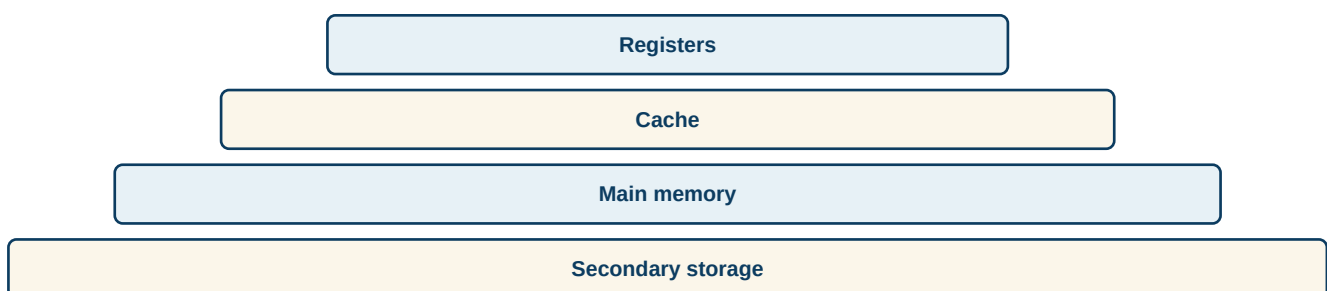


Figure 2.1 Memory ladder. The top is the fastest and the smallest. The bottom is the slowest and the largest.

2.3 Secondary memory

Secondary memory. Non-volatile storage that keeps programs and data for later use. It is larger and slower than main memory. The CPU uses its contents after they are loaded into RAM.

- A **hard disk drive** stores data on magnetic platters. A head reads and writes while the platter spins. Capacity is large. It is the traditional store inside a desktop system unit.
- A **USB flash drive**, also called a pen drive or data traveller, uses flash memory. It is portable, has no spinning part, and plugs into a USB port.
- An **external hard drive** is a hard disk in its own case, usually attached by USB, used for backup and for moving large folders.
- An **SD card** or **microSD card** is a flash card used in phones, cameras and some laptops.
- A **combo drive** is an optical drive that works with more than one disc type, commonly CD and DVD. Discs are removable. CD-R records once. CD-RW can be rewritten.

Choose the device from the job. A running program needs RAM. A file you will open next month needs secondary storage. A copy you will carry to college needs a USB drive or a memory card. A full backup of a lab machine needs an external hard drive or a large internal disk.

Exam point

Do not call a hard disk ROM. Both keep data without power. ROM is primary semiconductor memory for permanent instructions. A hard disk is secondary magnetic storage for files.

Key terms

Term	Meaning to write in an answer
Bit	A single binary digit, 0 or 1.
Nibble	Four bits.
Byte	Eight bits.
RAM	Volatile working memory with direct access to any location.
SRAM	Fast RAM that stores bits in flip-flops and needs no refresh.
DRAM	RAM that stores bits as charge and needs refresh.
ROM	Non-volatile memory read in ordinary operation.
EEPROM	ROM that can be erased and rewritten electrically.
Cache	Small, fast SRAM used to feed the processor.

Bring it together

- Measure memory from the bit up. In this course 1 KB is 1024 bytes.
- RAM is volatile working memory. SRAM is cache. DRAM is main memory.
- ROM, PROM, EPROM and EEPROM are non-volatile. They differ in how they are programmed and erased.
- Hard disks, flash drives, memory cards and optical combo drives are secondary storage.

Do this in the practical

- 1 Convert 3 KB to bytes and to bits on paper. Then read the size of a real file in the operating system and say whether it is closer to a KB, an MB or a GB.

Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

1. A nibble contains:

- A. 2 bits
- B. 4 bits
- C. 8 bits
- D. 1024 bits

2. How many bytes are there in 1 KB in this course?

- A. 1000
- B. 1024
- C. 8
- D. 512

3. Which memory needs to be refreshed?

- A. SRAM
- B. DRAM
- C. DVD
- D. PROM

4. EPROM is erased by:

- A. Ultraviolet light
- B. A magnet
- C. A paper jam
- D. A plotter

5. Which memory is typically used as cache?

- A. DRAM
- B. SRAM
- C. CD-R
- D. Blu-ray

6. A pen drive is:

- A. Volatile primary memory
- B. Secondary flash storage
- C. A language translator
- D. An input-only device

7. PROM can be programmed:

- A. Every time the PC boots
- B. Once by the user
- C. Only by ultraviolet light
- D. Only by a router

8. Which statement about ROM is correct?

- A. It is volatile
- B. It commonly holds start-up instructions
- C. It is a magnetic platter
- D. It is erased by deleting a folder

Answer key

Q	Answer	Why this is the answer
1	B	A nibble is half a byte, so 4 bits.
2	B	The course uses 1024, which is 2 to the power 10.
3	B	DRAM stores charge in capacitors, and that charge must be refreshed.
4	A	EPROM is erased by ultraviolet light shone through its window.
5	B	SRAM is fast enough and costly enough that it is used in small cache.
6	B	A pen drive keeps files without power and uses flash memory.
7	B	PROM is programmable once. After programming it stays fixed.
8	B	ROM is non-volatile and is used for permanent start-up code.

Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

Q1. Convert 4 MB into KB.

Model answer

1 MB = 1024 KB, so 4 MB = 4 x 1024 = 4096 KB.

Q2. Differentiate SRAM and DRAM.

Model answer

SRAM stores each bit in a transistor flip-flop, does not need refresh, and is faster and costlier. It is used as cache.
DRAM stores each bit as a charge, needs refresh, and is cheaper per bit. It is used as main memory.

Q3. Differentiate EPROM and EEPROM.

Model answer

EPROM is erased by ultraviolet light and is then reprogrammed. EEPROM is erased and rewritten by electrical signals, so it can be updated in the circuit. Both are non-volatile programmable ROM.

Q4. Why is RAM called volatile?

Model answer

RAM needs electrical power to hold its contents. When power is removed, the stored program and data disappear.

Q5. What is a combo drive?

Model answer

A combo drive is an optical drive that reads or writes more than one disc type, commonly both CD and DVD.

Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

Question 1. Explain the units of memory from bit to terabyte, with one numerical example.

- 1 Define bit, nibble and byte with the numbers 1, 4 and 8.
- 2 Give the ladder KB, MB, GB and TB, each 1024 of the unit below.

- 3 Work one conversion in full, for example $2 \text{ KB} = 2048 \text{ bytes} = 16384 \text{ bits}$. Show the multiplication.
- 4 Say why a standard matters: file sizes and memory chips are compared with these units.

Question 2. Describe RAM and ROM, including SRAM, DRAM, PROM, EPROM and EEPROM.

- 1 Define RAM as volatile direct-access working memory, and ROM as non-volatile memory read during ordinary work.
- 2 Contrast SRAM and DRAM with cell, refresh, speed, cost and use.
- 3 Describe PROM once, EPROM with ultraviolet erase, and EEPROM with electrical erase.
- 4 Give the job of each: DRAM as main memory, SRAM as cache, ROM family as permanent or semi-permanent instructions and firmware.